

SUNDANCE 8 NNW, WY, USA

WMO: 755750

Lat: 44.5169N

Lon: 104.4363W

Elev: 1765

StdP: 81.84

Time Zone: -7.00 (NAM)

Period: 07-19

WBAN: 94088

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-24.1	-21.3	-27.2	0.4	-20.9	-24.7	0.5	-19.5	8.3	-1.1	7.6	-1.4	2.0	N/A	0.562

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	28.0	14.8	26.4	14.6	25.0	14.4	17.7	23.4	16.7	22.8	15.9	22.5	2.4	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.7	13.8	20.8	14.6	12.9	20.0	13.4	11.9	19.0	57.4	23.4	54.0	23.0	51.2	22.5	22.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.3	6.4	5.7	-27.9	30.4	3.3	1.5	-30.3	31.5	-32.2	32.4	-34.0	33.2	-36.4	34.2
			-27.9	19.6	3.3	1.3	-30.2	20.5	-32.2	21.2	-34.1	21.9	-36.5	22.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	5.1	-5.0	-6.3	-0.2	2.8	7.8	13.8	18.0	17.2	13.1	5.2	-0.2	-5.8	(6)
(7)		DBStd	10.31	6.72	6.91	6.53	5.57	5.37	4.35	3.47	3.85	5.44	6.28	6.72	7.12	(7)
(8)		HDD10.0	2590	466	456	320	226	107	13	1	2	32	171	309	488	(8)
(9)		HDD18.3	4927	724	689	576	467	328	147	48	68	170	408	556	747	(9)
(10)		CDD10.0	803	0	0	3	8	39	127	249	225	124	23	4	0	(10)
(11)		CDD18.3	97	0	0	0	0	2	11	38	33	13	1	0	0	(11)
(12)		CDH23.3	711	0	0	0	0	8	85	291	246	78	3	0	0	(12)
(13)		CDH26.7	121	0	0	0	0	1	16	56	42	6	0	0	0	(13)
(14)	Wind	WSAvg	2.7	3.3	3.0	2.9	2.8	2.6	2.4	2.0	2.1	2.4	2.7	3.0	3.1	(14)
(15)	Precipitation	PrecAvg	469	16	17	27	55	79	76	54	39	30	41	21	15	(15)
(16)		PrecMax	613	37	35	56	94	166	171	95	77	78	134	76	42	(16)
(17)		PrecMin	344	5	6	13	24	31	15	16	12	10	7	2	2	(17)
(18)		PrecStd	80	9	7	11	20	38	40	22	19	17	33	15	10	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.2	10.6	17.1	20.3	24.5	29.1	30.6	29.8	27.6	22.5	16.2	10.0	(19)
(20)		MCWB	3.6	3.3	6.5	8.7	11.8	14.2	14.9	14.6	14.0	10.7	6.2	3.2	3.2	(20)
(21)		2%	DB	7.2	8.1	14.1	16.7	20.8	25.5	28.2	27.9	25.6	19.1	13.5	7.4	(21)
(22)		MCWB	1.2	2.3	5.4	7.6	10.9	14.5	15.5	14.1	13.2	8.8	5.1	1.6	1.6	(22)
(23)		5%	DB	5.5	5.7	11.2	13.8	18.8	23.7	26.5	26.0	23.8	16.9	11.4	5.4	(23)
(24)		MCWB	0.3	1.0	3.9	6.4	10.9	14.3	15.4	14.2	12.2	8.0	4.1	0.3	0.3	(24)
(25)		10%	DB	3.9	3.3	8.9	11.3	16.4	21.7	24.8	24.4	21.8	14.6	9.3	3.6	(25)
(26)		MCWB	-0.6	-0.5	2.8	5.1	9.6	13.7	15.3	14.0	11.3	7.1	3.0	-0.7	-0.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.7	4.1	7.7	10.0	14.3	18.7	19.3	17.7	17.5	11.6	6.8	3.9	(27)
(28)		MCDWB	11.0	9.0	16.0	18.3	19.6	24.3	24.9	23.2	23.4	19.6	15.4	9.6	9.6	(28)
(29)		2%	WB	1.8	2.8	5.9	8.3	12.8	16.8	18.0	16.7	15.2	10.2	5.3	2.1	(29)
(30)		MCDWB	6.4	8.0	14.1	15.9	17.9	22.4	23.6	22.7	21.5	17.1	12.5	6.6	6.6	(30)
(31)		5%	WB	0.7	1.4	4.4	7.0	11.6	15.5	17.2	15.9	13.7	8.9	4.3	0.8	(31)
(32)		MCDWB	4.9	6.0	11.0	13.6	17.1	21.8	23.1	22.5	20.9	15.7	10.8	5.0	5.0	(32)
(33)		10%	WB	-0.3	0.0	3.2	5.5	10.3	14.5	16.3	15.1	12.6	7.6	3.0	-0.3	(33)
(34)		MCDWB	3.6	3.8	8.8	11.1	15.9	20.9	22.8	22.3	19.9	13.7	8.6	4.1	4.1	(34)
(35)	Mean Daily Temperature Range	MDBR	7.9	8.3	8.4	8.9	8.5	10.3	10.9	11.0	10.5	8.6	8.0	7.5	(35)	
(36)		5% DB	MCDBR	8.7	8.3	10.1	12.1	10.5	12.5	12.1	12.2	11.6	10.9	9.4	8.0	(36)
(37)		MCWBR	5.4	4.9	5.4	6.3	5.6	6.5	5.7	5.5	5.0	5.1	4.9	4.6	(37)	
(38)		5% WB	MCDBR	9.4	9.1	10.1	11.8	9.9	11.8	10.9	11.1	10.5	10.5	8.8	8.0	(38)
(39)		MCWBR	6.4	5.6	5.7	6.7	5.7	7.1	6.6	6.0	5.1	5.8	4.9	5.5	5.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.234	0.241	0.257	0.279	0.296	0.306	0.324	0.325	0.293	0.268	0.244	0.231	(40)	
(41)		taud	2.609	2.596	2.605	2.532	2.550	2.542	2.508	2.485	2.573	2.619	2.598	2.606	(41)	
(42)		Ebn at Noon	932	980	997	989	974	960	938	928	940	930	908	903	(42)	
(43)		Edh at Noon	59	72	83	97	99	100	102	101	85	70	59	54	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.84	2.77	4.03	4.96	5.92	6.96	7.23	6.21	4.76	3.07	2.04	1.52	(44)	
(45)		RadStd	0.09	0.15	0.30	0.26	0.58	0.57	0.26	0.22	0.32	0.27	0.18	0.10	(45)	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page